

Theoretical study on ReefBooster stability



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Summary

The offshore connection of Hollandse Kust West Alpha and Hollandse Kust West Beta to Tennet's onshore grid in the Netherlands, requires three connection cables crossing several in-service cables and pipelines. Rock berms at these cable-crossing locations create the opportunity to improve local biodiversity by designing the berms in an eco-friendly way (Nature-Inclusive Design; NID). BlueLinked has been contracted by Boskalis and TenneT to perform a feasibility study for deploying the ReefBooster biodegradable tiles, with flat oyster spat, to kick start active restoration on these rock berms. In turn, BlueLinked has contracted Deltares to perform a desk study on the hydrodynamic stability and potential silt accumulation on the ReefBooster tiles for two (2) cable-crossing locations: a nearshore and an offshore location. Two types of rock berms are designed for two different cross sections along the rock berm, i.e., at the Cable Separation System (CSS) and at Cable-only.

The ReefBooster tile is schematized as equivalent to rock to perform the stability assessment, with an equivalent rock diameter of 0.046 m and densities of 2000 and 2330 kg/m³. The stability assessment of the tiles under extreme conditions results in the advice to use a larger density of 2330 kg/m³ (or more) for the ReefBooster tiles to avoid displacement or significant displacement. In case a lower density is used, only the offshore location would be suitable, though displacement is still likely to occur. The stability assessment of the tiles under normal conditions shows that, at the offshore location, the tiles remain stable. The lighter tiles are likely to show some rocking and displacement within a maximum distance of a few times the ReefBooster tile diameter, in unfavourable conditions. For the nearshore location, a heavier ReefBooster tile would be required to prevent displacement or significant displacement.

Silt accumulation on the ReefBooster tiles during a period of 10 years (from 2025 to 2035) appeared to be unlikely for both the nearshore and offshore location.

Placement of the ReefBooster tiles on the offshore cable crossing location shows a lower risk of displacement of the tiles, especially with a higher density of 2330 kg/m³ (or more). When placed here, on top of the CSS rock berms, even during a period of 20 years (2025 to 2045) siltation is very unlikely.

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1 Introduction

1.1 General introduction

TenneT is responsible for the realization and operation of the offshore connection of two 700 MW offshore platforms, namely Hollandse Kust West Alpha (hereafter HKWA) and Hollandse Kust West Beta (hereafter HKWB), to TenneT's onshore grid in the Netherlands.

Consortium Ningbo Orient Cable & Boskalis (NBOS), the Contractor, is responsible for the two 65-kilometre-long 220kV export cables (Sea Cables HKwB01 and HKwB02) and a 9-kilometre-long 66kV interconnector cable between the HKWA and HKWB (Interlink Sea Cable HKWBIL).

The three cables will cross several in-service cables and pipelines, requiring in total 17 crossing (rock berm) structures. Figure 1.1 presents the offshore platforms, export and interconnector cables, and the cable crossings.



Figure 1.1 Offshore platforms, export and interconnector cables and cable crossing locations of the HKWB project.

Seven (7) of these rock berms are identified by TenneT as potentially suitable for Nature Inclusive Design (NID), accounting for shipping intensity, asset owners and environmental conditions.

To that end, BlueLinked has been contacted by Boskalis and TenneT to perform a feasibility study for deploying the ReefBooster biodegradable tiles with flat oyster spat, as a kick starter for active restoration. In turn, BlueLinked has contacted Deltares for performing a desk study on the hydrodynamic stability and potential silt accumulation on the ReefBoosters.

1.2 Background

Large-scale restoration of the historically abundant and currently extinct oyster reefs in the North Sea is one of the focal points in the effort to achieve a healthy and biodiverse ecosystem. The ReefBooster developed by BlueLinked may contribute to this restoration effort. This

ReefBooster is a tile on which oyster spat may be cultivated in an onshore facility, then to be deployed in an offshore setting. The tile is made from materials that degrade within a few years from deployment. During these years, the oyster spat may gradually grow into an oyster reef given the right environmental boundary conditions, such as limited movement during storms. The installation process does not require heavy equipment, and there is no need for decommissioning. The tile is modular, meaning that its size and weight can vary depending on the application and stability requirements (Raspoort, 2022; Boskalis, 2024).



Figure 1.2 The ReefBooster tiles at different sizes (Raspoort, 2022).

For this specific project, the ReefBooster will be deployed at a minimum of one (1) and a maximum of six (6) rock berm structures, to be selected out of the seven (7) locations identified by TenneT as suitable for Nature-Inclusive Design (NID), see Section 2.1 for more background information). This selection will be based on the location-specific hydrodynamic and morphodynamic conditions that influence the stability of the tiles and the potential of silt accumulation on the reef tiles. The hydrodynamic and morphodynamic behaviour of the ReefBooster also depends on the exact placement of the tiles relative to the rock berm, e.g., crest, slopes, flat seabed. For the latter, this can be either on the adjacent sandy seabed or on the extended rock cover (see Figure 1.3).

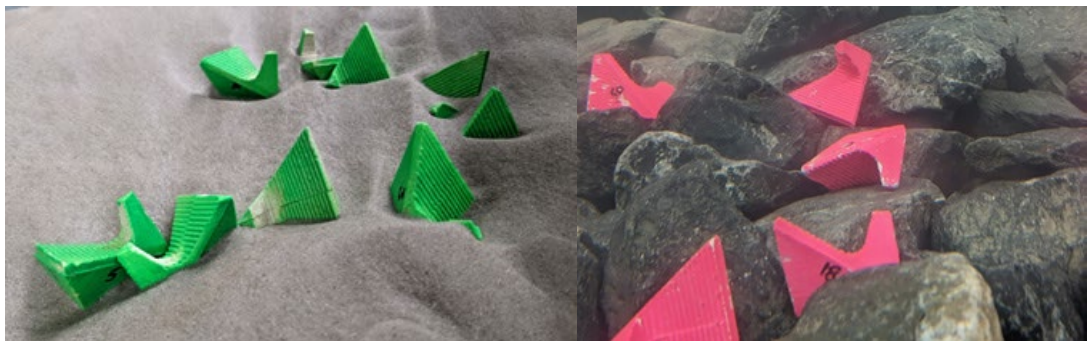


Figure 1.3 ReefBooster tiles situated directly on the seabed (left) and on loose rock substrate (right) (Raspoort, 2022).

In theory, to address both risks of instability and siltation, a larger size from the available range in dimensions of the ReefBooster (7-30 cm) can be selected for deployment, depending on the environmental conditions, permits and the rock berm design, amongst other factors. For this project however, the choice of the maximum tile size is limited by the rock size of the underlying sprinkle layer (8 cm), which is required as a topping layer at all rock berm structures (all falling outside the 500 m safety zone) to smoothen the shape and hence mitigate the risk that fishing gear and other dragged objects will get damaged (and vice versa).

1.3 Research objective

From a technical perspective, the main uncertainty concerning the deployment of the ReefBooster tiles at the cable crossings of HKWB, relates to the expected hydraulic behaviour. This involves two main aspects: (1) hydrodynamic stability and (2) potential siltation.

Deltares will perform a desk study, which consists of the following activities:

1. Schematize the ReefBooster into a scour protection equivalent (i.e. a rock or similar object), so it can be evaluated using existing design criteria.
2. Perform a calculation on ReefBooster stability in representative offshore conditions using design criteria and expert judgment, based on conditions for two (2) offshore sites.
3. Advise on possibility of silt accumulation on ReefBoosters, depending on location on the rock berm, for two (2) offshore sites.

As this is a first step in exploring the deployment of ReefBoosters in an offshore setting, the approach and results of this desk study are exploratory. This means the results should be perceived as indicative and not conclusive for the hydraulic performance of the ReefBooster. Further detailing of calculations and assumptions, as well as verification of the calculations using physical model testing, are recommended next steps in further development of the ReefBooster.

1.4 Readers guide

Chapter 2 describes the boundary conditions for the stability and siltation assessment. It includes the rock berm characteristics, ReefBooster characteristics, metocean conditions and morphodynamic conditions at the specific cable crossing locations. Chapter 3 presents the results of the stability and siltation analysis, on which Chapter 4 concludes.

2 Boundary conditions

2.1 Cable crossing locations

The two (2) 220 kV export cables (Sea Cables HKWB01 and HKWB02) cross in-service cables at 5 locations and in-service pipelines at 3 locations. Hence, for the export cables 16 crossing structures are required in total. The HKWBIL Interlink Sea Cable crosses one (1) in-service pipeline, thus requiring one (1) crossing structure on that route.

Out of these crossings, TenneT has selected seven (7) locations that are suitable for NID, based on several considerations. These are presented in Table 2.1.

To narrow this down to two locations, we consider that wave-induced forcing is often the dominant factor for the stability of rocks and structures in an offshore environment. Generally, waves penetrate further into the water column at shallower locations. Hence, wave forcing near the seabed increases with a decreasing water depth (for the same wave conditions).

Therefore, the seven (7) locations are categorized in two (2) clusters, based on the local water depth (Figure 2.1). These nearshore and offshore clusters have a seabed level that varies in the range of -17.7 to -20.8 m LAT and -24.4 to 29.0 m LAT, respectively. A single location from each of the two clusters is selected, which is globally representative for metocean conditions and water depth in that cluster.

Based on preference of the client, location 08 (Figure 2.1) was selected from the nearshore cluster, while from the offshore cluster location 16 (Figure 2.1) was selected. Hereafter, these are addressed as the nearshore and offshore location, respectively.

Table 2.1 Crossing locations that are pre-selected by TenneT as potentially suitable for NID. Seabed levels are from (DHI, 2020).

Crossing ID	Crossed Asset	Easting [m, UTM 31 N]	Northing [m, UTM 31 N]	Seabed level [m LAT]
06 - KB0074	Atlantic Segment 1 B1	597023.997	5823435.304	-17.68
08 - KB0026(1)	Atlantic Segment 1 B2 (A)	595928.318	5824686.714	-18.87
10 - KB0026(2)	Atlantic Segment 1 B2 (B)	588965.870	5831592.107	-20.75
12 - KB0029	PANGEA Segment 2	563710.749	5829598.502	-24.36
14 - PL0207_PR	P9-BP6-D	556751.451	5829772.551	-25.38
16 - PL0109_PR	P9 Horizon AQ1 Helder AW	554469.452	5829562.550	-27.80
17 - PL0207_PR	P9-BP6-D (Interlink cable)	552634.999	5834531.176	-29.06



Figure 2.1 Map of the cable crossings along the HKW-B cables, with the two selected cable crossing locations from the nearshore and offshore cluster of cable crossings (Tennet, 2024).

2.2 Rock berm characteristics

The design life of the rock berms is indicated to be 40 years. To cope with the hydrodynamic loading over this period, two different rock berms have been designed by Boskalis for the seven (7) cable crossing locations (Tennet, 2024). A double-layer system comprising of a separate filter (with an OSG 22/80mm rock grading) and armour layer (with a OWG 5-40kg rock grading) is required at the two (2) shallowest cable crossings. The remaining five (5) crossings will be protected by means of a single-layer system with an OSG 90/250mm rock grading.

All cable crossings fall outside the 500 m safety zone and hence the Dutch authorities (Rijkswaterstaat) require that a Sprinkle Layer of rock with a minimum thickness of 0.2 m after installation and a $D_{90} \leq 80$ mm (to be supplied with the OSG 22/80mm grading, as in the filter of the double-layer system) is added on top of the rock berms, to mitigate risks in case of dropping objects. The ReefBooster tiles will be deployed on top of this Sprinkle Layer.

The two rock berm designs, along with the Sprinkle Layer are presented in in Figure 2.2 for two different cross sections along the rock berm, i.e., at the Cable Separation System (CSS) and at Cable-only. The relevant rock berm (external) dimensions are summarized in Table 2.2.

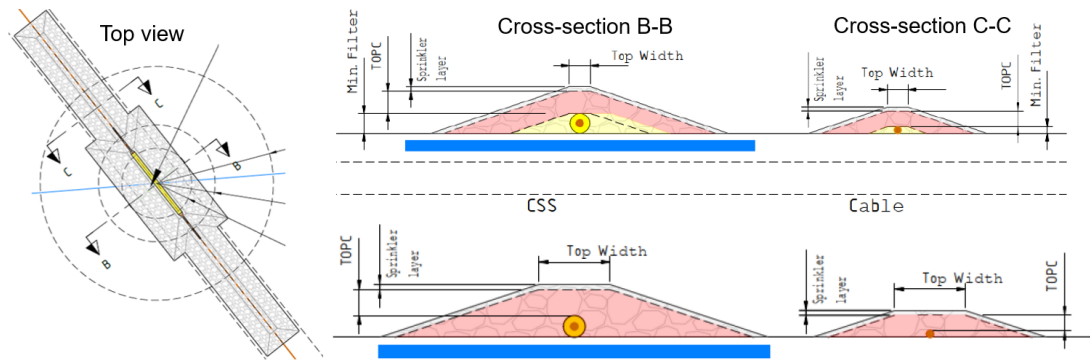


Figure 2.2 Top view of a typical rock berm at the HKW (left panel). Double- and single-layer system (top and bottom panels to the right respectively), both of which are topped with a sprinkle layer on top of which the ReefBoosters will be deployed (Tennet, 2024).

Table 2.2 Rock berm characteristics at the two selected crossing locations. The total protection height (H_{prot}) also accounts for the (minimum) Sprinkle Layer of 0.2 m after installation.

Crossing ID	Section	D [m]	Top Width, W_{top} [m]	Side slope, $\cot(\alpha)$	Top Cover, TOPC [m]	Berm Height, h_{berm} [m]	Total Protection Height, H_{prot} [m]	Grading (filter / armour)
08 KB0026(1)	CSS	0.92	1		1.05	1.97	2.17	OSG 22/80mm / OWG 5-40kg
	Cable	0.28	1		0.75	1.03	1.23	
16 PL0109_PR	CSS	0.92	1		1.00	1.92	2.12	OSG 90/250mm
	Cable	0.28	1		0.65	0.93	1.13	

Table 2.2 shows that when ReefBoosters are deployed on the crest of the rock berms, they will be situated on a sill, with a height ranging between 1.13 m and 1.23 m (Cable-only section) and between 2.12 and 2.17 m (CSS section) depending on the cable crossing location (minimum design). This means that the tiles will be exposed to amplified hydrodynamic loads compared to the seabed, which needs to be accounted for in the stability assessment.

2.3 ReefBooster characteristics

For this project, ReefBooster dimensions are selected to be equally large to the sprinkle layer rocks ($D_{90} \leq 80$ mm, see Section 2.2). As a result, they are characterized by a width of 80mm and two sides of 54mm (Figure 2.3). This results in a volume of solids of 51ml (0.000051 m³). The density varies between 2000-2330 kg/m³. The foreseen spreading density is 10 units per m².

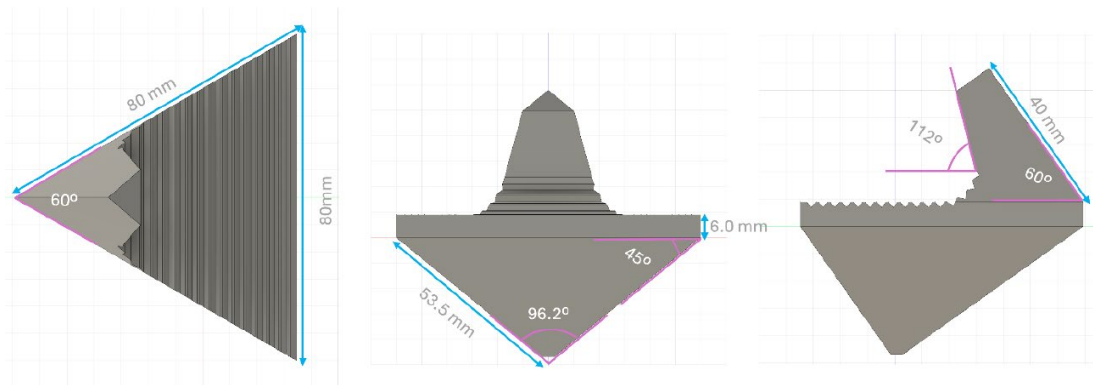


Figure 2.3 Design drawing of the ReefBooster from different sides (from left to right: top, bottom and side).

The ReefBooster units are created from a naturally degradable material with a design life of 5 to 10 years. However, the oysters themselves can remain on the rock berm structure for much longer, as they attach themselves to the hard substrate of the rock berm.

2.4 Metocean conditions

Based on the lifetime and the available metocean data, a return period (RP) of 50 years has been determined by the Client as the normative (storm) condition for the assessment of the stability of the ReefBooster. Subsequently, extreme metocean conditions concerning waves, currents and water levels derived from (TenneT, 2024; DHI, 2020) are provided from the Client to Deltares, for the two (2) crossing locations, corresponding to return period of 50 (RP50) and 10 years (RP10).

Table 2.3 Extreme metocean conditions at the two (2) cable crossing locations (during High Water and Low Water). Current speeds are independent values and not associated with the H_{m0} .

Crossing ID	RP10				RP50			
	LW/HW H_{m0} [m]	T_p [s]	LW/HW WL [m LAT]	U_c [m/s]	LW/HW H_{m0} [m]	T_p [s]	LW/HW WL [m LAT]	U_c [m/s]
08 - KB0026(1)	4.92 / 6.44	13.6	-0.96 / 1.71	1.13	5.28 / 6.83	14.0	-1.17 / 1.51	1.18
16 - PL0109_PR	5.71 / 7.06	13.6	-0.93 / 2.00	1.14	6.22 / 7.63	14.1	-1.12 / 2.30	1.19

Additionally, to perform an analysis of ReefBooster stability under year-round conditions, we extracted the associated (hourly) timeseries of the full hindcast period (1979-2023) of the same metocean study (TenneT, 2024; DHI, 2020). Figure 2.4 and Figure 2.5 show the full hindcast period for the nearshore and offshore location respectively, retrieved from (DHI, 2024)

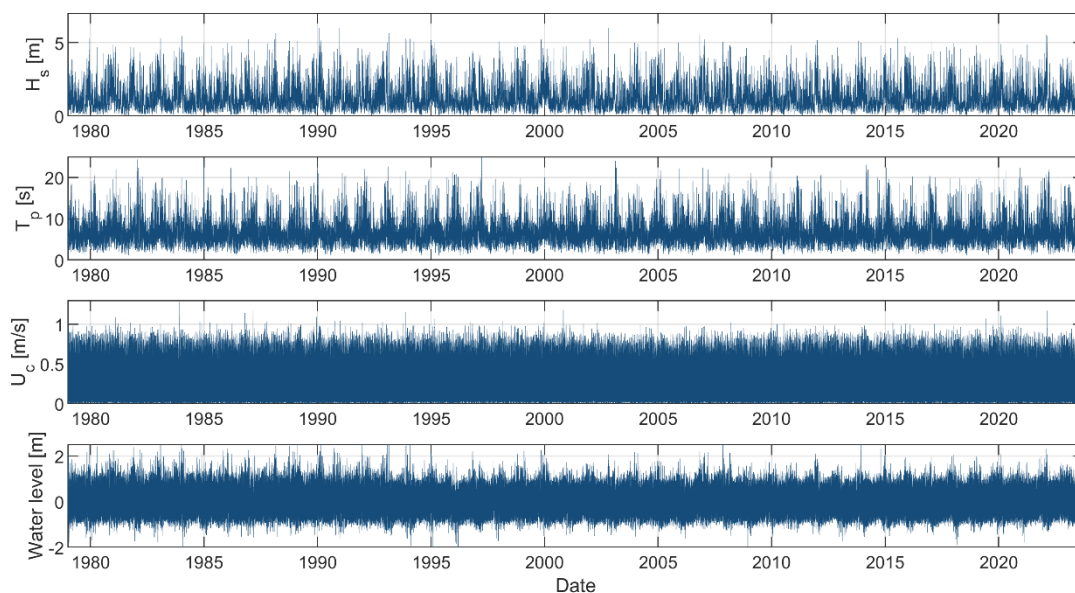


Figure 2.4 Extracted timeseries of the metocean conditions at the nearshore location.

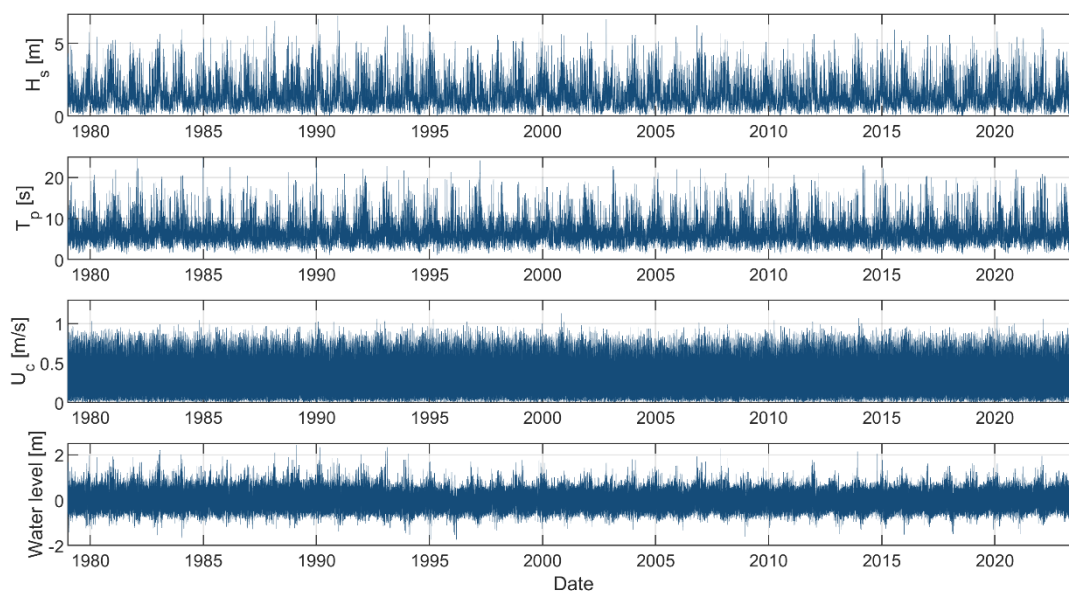


Figure 2.5 Extracted timeseries of the metocean conditions at the offshore location.

2.5 Morphodynamic conditions

Both the seabed level at the nearshore and offshore location are checked with available data from recent bathymetric surveys. The nearshore location aligns very well with the seabed level described by DHI, (2020), but the offshore location is in an area with sandwaves for which the grid resolution in DHI, (2020) is too coarse and does not adequately capture sandwave dynamics.

The bedlevel from the bathymetric surveys is more accurate and is therefore used in this study. The adapted seabed level for the offshore location is included in Table 2.1. Some uncertainty in the bed level measurements and the extrapolation of the measurements to the current state of the bed level is included in the Monte Carlo simulation (see Section 3.3.2) and is as follows:

- Nearshore location (location 08): [-0.3 0.3] m
- Offshore location (location 16): [-0.5 0.8] m

3 Analysis and results

3.1 Schematizing the ReefBooster into an equivalent rock

The ReefBooster dimensions are fixed and correspond to a volume of 51ml. Two densities are considered in this study, as the tile density can range between 2000 to 2330 kg/m³.

To evaluate their stability using existing design criteria we need to schematise them as an equivalent rock with a representative diameter and density. The equivalent rock diameter is the diameter of a sphere with the same weight and density as the corresponding tile. To calculate the density of the equivalent sphere, the weight is kept equal to the weight of the ReefBooster tile, and the volume of the equivalent sphere is used. This results in the two equivalent rock cases listed in Table 3.1. Both cases will be included in the stability analysis. The equivalent rock diameter will be directly used in the calculations and is not converted to a D_{n50} .

Table 3.1 Equivalent rock characteristics.

ReefBooster density [kg/m ³]	Reef booster volume [m ³]	Reef booster weight [kg]	Equivalent rock diameter [m]
2000	0.000051	0.102	0.046
2330	0.000051	0.119	0.046

3.2 Method for assessing ReefBooster stability

We use two methods for assessing ReefBooster stability. Both use the mobility number to evaluate stability:

1. Compute the mobility parameter for extreme conditions, i.e., storms with a return period of 10 or 50 years.
2. Compute the mobility parameter for normal conditions by performing a (probabilistic) Monte Carlo analysis using hindcast timeseries.

The HasPro scour handbook (Deltares, 2023) describes the mobility number (MOB) as the undisturbed relative mobility value of rock, under hydrodynamic loading of combined waves and currents. It is given by the ratio between the Shields number (θ) and the critical Shields number (θ_{cr}) of a certain equivalent rock size. The input parameters for the calculations are based on the relevant boundary conditions:

- Hydrodynamic conditions, i.e.:
 - o Significant wave height (H_s)
 - o Wave period (T_p)
 - o Flow velocity (U_c)
 - o Water level (WL)
- Equivalent rock characteristics, i.e.:
 - o Equivalent rock diameter (D_{50})
 - o Equivalent rock density (ρ)

The critical values for the MOB parameter are as follows and lead to the criteria in Table 3.2:

- MOB > 0.95 (significant) motion: likely rocking and displacement within a distance of a few times the ReefBooster tile diameter
- MOB > 1.2 displacement: likely displacement within the footprint of the cable crossing
- MOB > 1.5 significant displacement: likely displacement outside the footprint of the cable crossing

The current values are taken based on expert judgement and on interpretation of rock berm stability tests performed during JIP HaSPro (Deltares, 2023). These values should be considered indicative for the present stability analysis, as they could be material dependent. To derive exact mobility limits for the ReefBooster tiles, physical scale model testing is recommended.

3.3 ReefBooster stability in waves and currents

3.3.1 Stability in extreme conditions

Two load cases are considered in this study: a RP10 storm and a RP50 storm (Section 2.4). For each load case, we performed a stability analysis of two equivalent rock sizes at three different sill heights (on the seabed, on top of the Cable-only berm, and the CSS rock berm), for both the offshore and the nearshore location. The parameters are based on the characteristics and conditions described in Chapter 2.

Results

Table 3.3 and Table 3.4 show the results of the extreme value analysis for the nearshore and offshore location, respectively. The different mobility limits indicated using different colours which are defined in Table 3.2.

Table 3.2 Colours indicating the degree of movement of the ReefBooster tiles under extreme conditions.

	MOB < 0.95	No motion	Likely no motion or limited rocking
	$0.95 \leq \text{MOB} < 1.20$	(Significant) motion	Likely rocking and displacement within a distance of a few times the ReefBooster tile diameter
	$1.20 \leq \text{MOB} < 1.50$	Displacement	Likely displacement within the footprint of the cable crossing
	MOB ≥ 1.5	Significant displacement	Likely displacement outside the footprint of the cable crossing

Table 3.3 Nearshore location (location 08). The calculated mobility parameter for the two equivalent rocks, two load cases, three sill heights and two design water levels.

Density case	RP	Berm height	Design water level	Mobility parameter
2000	10	Seabed	LW	1.45
			HW	1.82
		Cable	LW	1.61
			HW	2.01
		CSS	LW	1.76
			HW	2.18
	50	Seabed	LW	1.63
			HW	1.99
		Cable	LW	1.81
			HW	2.20
		CSS	LW	1.98
			HW	2.38
2330	10	Seabed	LW	1.08
			HW	1.36
		Cable	LW	1.21
			HW	1.50
		CSS	LW	1.32
			HW	1.63
	50	Seabed	LW	1.22
			HW	1.49
		Cable	LW	1.35
			HW	1.64
		CSS	LW	1.48
			HW	1.78

Table 3.4 Offshore location (location 16). The calculated mobility parameter for the two load cases, three sill heights and two design water levels.

Density case	RP	Berm height	Design water level	Mobility parameter
2000	10	Seabed	LW	1.05
			HW	1.31
		Cable	LW	1.13
			HW	1.41
		CSS	LW	1.20
			HW	1.50
	50	Seabed	LW	1.18
			HW	1.46
		Cable	LW	1.27
			HW	1.56
		CSS	LW	1.35
			HW	1.66
2330	10	Seabed	LW	0.78
			HW	0.98
		Cable	LW	0.84
			HW	1.05
		CSS	LW	0.90
			HW	1.12
	50	Seabed	LW	0.88
			HW	1.09
		Cable	LW	0.95
			HW	1.17
		CSS	LW	1.01
			HW	1.24

Discussion

The results for the stability at the nearshore location show very mobile ReefBooster tiles if their density is low. Significant displacement of lighter tiles is likely here. Increasing the density of the tiles enhances their stability and with (significant) motion or displacement in mind, the tiles could be placed on the Cable-only rock berm, with some risk of significant displacement due to a RP50 storm event. This, however, depends on what is acceptable motion of the tiles from an ecological perspective, i.e. whether motion of the tiles or some displacement still enables proper reef development. Further development of the ReefBooster should include this consideration. Placing the tiles on the CSS rock berm likely results in significant displacement, as a RP10 storm event triggers significant displacement of the tiles and would therefore not be a suitable location.

For the offshore location, however, the heavier tiles with a density of 2330 kg/m³ are stable or show (significant) motion when placed on the Cable-only rock berm, both under a RP10 and RP50 storm event, but no displacement or significant displacement. Only when placed on the CSS rock berm displacement can occur under a RP50 storm event. Lighter tiles are more mobile and are on either the Cable-only rock berm or the CSS rock berm likely to be displaced during a RP10 storm event or significantly displaced under a RP50 storm event.

The situation where heavier tiles are placed on the rock berms at the offshore location is the only situation that results in stable to moving tiles, with no displacement.

3.3.2 Stability in normal conditions: Probabilistic Monte Carlo assessment

For both locations the mobility parameter of the ReefBooster tiles at three rock berm heights (seabed, cable-only and CSS) and two density cases is analysed for a one year metocean timeseries. This gives the stability under normal conditions.

Since the metocean conditions vary each year, the morphodynamic conditions have an uncertainty band, and the characteristics of the ReefBooster tiles and rock berm are not fixed yet or uncertain, a Monte Carlo (MC) simulation fits well to describe the probability of movement. Table 3.5 and Table 3.6 present the static and dynamic input parameters for the MC simulation. The number of randomly generated 'inputs' is 10.000 for each location and density case. A randomly generated input is drawn from each value range shown in Table 3.6. For example, for the metocean conditions, every realization draws randomly one full year between 1980 – 2022, containing a yearly timeseries of metocean conditions for the drawn year. In case of the installation tolerance a random generated value between 0 – 0.3 is added to the total height of the rock berm.

Table 3.5 Static input parameters for the Monte Carlo simulation.

Static input	Value
Density of seawater	1025 kg/m ³
Angle between waves and currents	0°
Roughness factor	2.5

Table 3.6 Dynamic input parameters for the Monte Carlo simulation.

Dynamic input	Distribution values/limits	Distribution
Yearly metocean timeseries (H_s , T_p , U_c and WL)	One full year between [1980 – 2022]	Uniform
Seabed level change	Nearshore location: mean = 0; std = 0.1 Offshore location: mean = -0.15; std = 0.2	Normal
Installation tolerance	[0 0.3] m for all rock berms at both locations	Uniform
Diameter equivalent rock	Mean = 0.046; std = 0.003	Normal
Density equivalent rock	Lighter: mean = 2000; std = 33 Heavier: mean = 2330; std = 39	Normal

The possible interlocking of the tiles with the sprinkle layer would increase the stability of the tiles, expressed by a higher critical shields parameter corresponding to the equivalent rock. The interlocking of the tiles is not directly considered in this study by increasing the critical shields parameter, but the mobility limits from (Deltares, 2023) are based on physical scale model tests with a rock berm where the rocks are interlocking.

Results

Figure 3.1 shows the cumulative hours in a year that the mobility parameter is larger than 0.95 ((significant) motion of the tiles) at the nearshore location, on the Cable-only rock berm and a ReefBooster tile with a density of 2000 kg/m³. The lower red line represents the 1th percentile and the upper red line the 99th percentile. The green line is the 50th percentile. To properly link the exceedance of the mobility parameter to the performance of the tiles (i.e.

whether reef development can occur) a follow up study is recommended, which would include the acceptability of movement of the tiles related to ecological performance.

Appendix A includes the cumulative hours for each location, berm height and density. Table 3.7 summarizes these results with the lower, median and upper bound representing the 1th, 50th and 99th percentile.

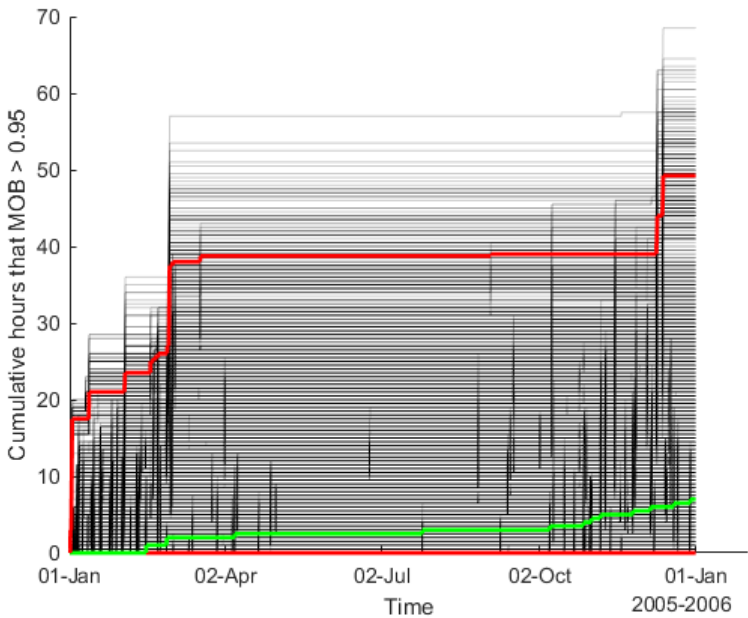


Figure 3.1 Density 2000 kg/m³, nearshore location (location 08) at the Cable-only rock berm.

Table 3.7 Nearshore location (location 08). The number of hours per year that the MOB parameter is larger than 0.95, 1.20 and 1.50. and thus indicates the mobility of the ReefBooster tiles.

Density case	Berm height	Hours per year that MOB > 0.95			Hours per year that MOB > 1.20			Hours per year that MOB > 1.50		
		Lower	Median	Upper	Lower	Median	Upper	Lower	Median	Upper
2000	Seabed	0	1	19.5	0	0	5.5	0	0	0
	Cable	0	7	50.5	0	0	11.5	0	0	1
	CSS	0	15.5	69	0	0	17	0	0	5
2330	Seabed	0	0	3	0	0	0	0	0	0
	Cable	0	0	10	0	0	0	0	0	0
	CSS	0	0	13	0	0	1.5	0	0	0

Table 3.8 Offshore location (location 16). The number of days per year that the MOB parameter is larger than 0.95, 1.20 and 1.50 and thus indicates the mobility of the ReefBooster tiles.

Density case	Berm height	Hours per year that MOB > 0.95			Hours per year that MOB > 1.20			Hours per year that MOB > 1.50		
		Lower	Median	Upper	Lower	Median	Upper	Lower	Median	Upper
2000	Seabed	0	0	4	0	0	0	0	0	0
	Cable	0	0	7.5	0	0	0	0	0	0
	CSS	0	0	10	0	0	0	0	0	0
2330	Seabed	0	0	0	0	0	0	0	0	0
	Cable	0	0	0	0	0	0	0	0	0
	CSS	0	0	0	0	0	0	0	0	0

Discussion

Under normal metocean conditions, placement of the lighter or heavier tiles at the offshore location (at all berm heights) shows stable tiles. A combination of stronger metocean conditions, a higher rock berm or a seabed level drop (unfavourable conditions) can cause (significant) motion of the lighter tiles at the Cable-only and CSS rock berm.

Placement of the heavier tiles at the nearshore location will result in stable tiles, with some probability of moving tiles (rocking and displacement within a distance of a few times the ReefBooster diameter) under unfavourable conditions. Lighter tiles placed at the Cable-only and CSS rock berms at the nearshore location might show (significant) motion of 7 to 16 hours (0.5 to 1.5 days) under expected conditions. With possible unfavourable conditions the tiles can likely show displacement over 50 to 69 hours (4 to 6 days) per year.

3.4 Possibility of silt accumulation on ReefBoosters

This section describes the details on the expected morphodynamic change at the two locations.

The offshore location is close to HKW-B (see Figure 3.2), for which we have morphodynamic data available at a relatively high temporal resolution, resulting in the availability of data with a higher time resolution here. The expected bed level change can be described well, within a normal uncertainty range. The nearshore location is limited in the availability of data, however, because the area is less dynamic (outside of sand wave area) the expected bed level change will be much lower.

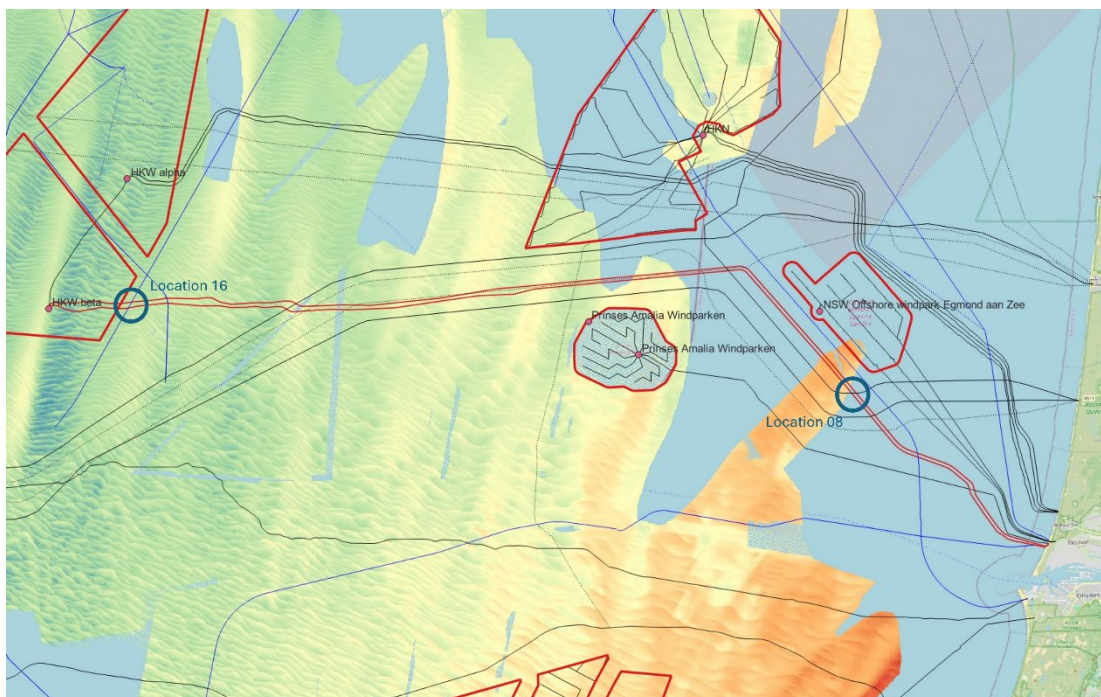


Figure 3.2 Bathymetry where sand waves are present. The nearshore and offshore location are marked in blue. Thick red lines indicate wind farms, the two thin red lines represent the cable from HKWB to land and the black thin lines show used or unused pipelines and telephone lines.

Nearshore location (location 08): Little morphodynamic change is expected for this cable crossing location, as sand waves are not regularly found in this area. Bed level data from 2001-2014 show bed level change ranging from 0 to 0.2 m. Megaripples may be present in this area, adding a 0.1 m band to this range. If we also consider measurement uncertainty, we will end up with a total range of -0.5 to 0.5 m of total bed level change over a timespan of around 10 years. This leads to a very low siltation probability, both on the cable (height = 1.03 m) as well as on the CSS (height = 1.97 m) rock berm.

Offshore location (location 16): The offshore cluster is in a dynamic sand wave area, and this specific location is situated in the through of a sand wave that migrates towards the East (towards the shore). Figure 3.3 presents the expected bed level in m+LAT for the offshore location up to 2060. Here, the Best Estimated Bathymetry (BEB) after 10 years (2025 – 2035) indicates a change in bed level of approximately +0.5m. The uncertainty envelopes show a range of uncertainty in the bed level change of approximately +0.3m to +1.0m. The height of the scour protection for the Cable-only and CSS at the offshore location are, respectively, 1.13m and 2.12m. Siltation during a 10-year period (2025 – 2035) is unlikely for both the Cable-only and the CSS rock berms.

Discussion

Over a period of 10 years (2025 – 2035), siltation at both the nearshore location (location 08) and offshore location (location 16) is unlikely.

An additional remark for the offshore location is that beyond the lifetime of the tiles themselves, oyster reefs have a tolerance limit for sedimentation. Figure 3.3 shows that a bed level change of +2.0 m is likely after 20 years (2025 – 2045), when an oyster reef might have developed. In this case, the silt accumulation on young oyster reefs could be avoided by placing the tiles on the CSS rock berm.

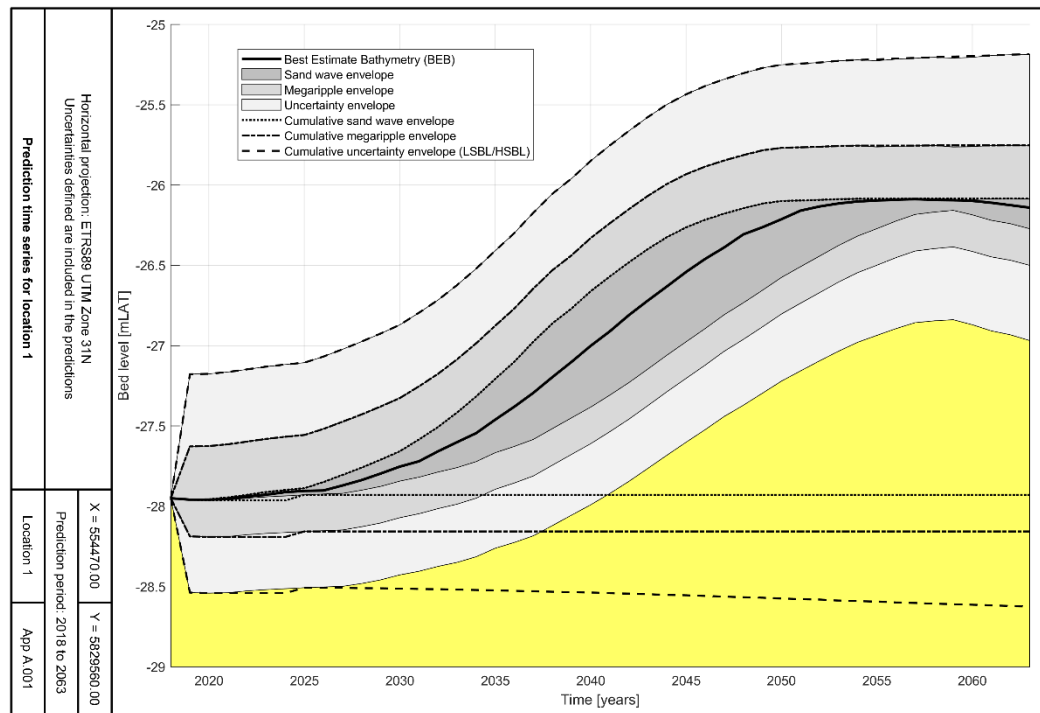


Figure 3.3 Expected bed level [m+LAT] for the offshore location up to 2060.

4 Conclusions and Recommendations

4.1 Conclusions

In this desk study, we assessed the hydrodynamic stability and possibility of silt accumulation on ReefBoosters in a marine environment. This study consisted of three main activities, for which we will draw the conclusions in this section:

1. Schematize the ReefBooster into a scour protection equivalent (i.e. a rock or similar object), so it can be evaluated using existing design criteria.
2. Perform a calculation on ReefBooster stability against waves and currents using design criteria and expert judgment, based on conditions for two offshore sites.
3. Advise on possibility of silt accumulation on ReefBoosters, depending on location on the rock berm, for two offshore sites.

4.1.1 Equivalent rock size

ReefBoosters with the same volume but different densities were evaluated by means of an equivalent rock size. The equivalent rock is characterized as a sphere and has the same weight as the represented ReefBooster tile, with a slightly different density and diameter (Table 4.1).

Table 4.1 Equivalent rock characteristics.

ReefBooster density [kg/m ³]	Reef booster volume [m ³]	Reef booster weight [kg]	Equivalent rock diameter [m]
2000	0.000051	0.102	0.046
2330	0.000051	0.119	0.046

4.1.2 ReefBooster stability in waves and currents

ReefBooster stability on the rock berm has been evaluated for two locations, using two different methods:

1. By computing the mobility parameter for extreme conditions, i.e., storms with a return period of 10 or 50 years.
2. Compute the mobility parameter for normal conditions by performing a (probabilistic) Monte Carlo analysis using hindcast timeseries.

Extreme conditions

ReefBoosters with a density of 2000 kg/m³ will be stable at:

- The offshore location (location 16), cable-only rock berm. Under RP10 conditions there will be motion or displacement, significant displacement only under RP50 conditions.

ReefBoosters with a density of 2330 kg/m³ will be stable at:

- Nearshore location, cable-only rock berm: Under RP10 conditions there will be (significant) motion or displacement, significant displacement only under RP50 storm conditions
- Offshore location, cable-only and CSS rock berm: Under RP10 conditions there will be (significant) motion but no displacement, displacement only at the CSS rock berm under RP50 conditions.

To avoid displacement, a larger density of 2330 kg/m³ (or more) is advised. In case a lower density is used, only the offshore location would be suitable, though (significant) motion and displacement can still be expected.

Normal conditions: Probabilistic Monte Carlo analysis

Complementary to the extreme conditions, the stability under normal conditions is assessed. At the offshore location, (significant) motion is expected for both the lighter and heavier ReefBoosters. (Significant) motion is expected for the lighter tiles, in case of unfavourable conditions. For the nearshore location, a heavier ReefBooster would be required to prevent displacement or significant displacement.

4.1.3 Possibility of silt accumulation

Sand waves are not regularly found at the nearshore location. Therefore, silt accumulation on the ReefBoosters due to morphodynamic changes in a 10-year period is unlikely.

The offshore location is situated in a sand wave field, more specifically in the trough of a sand wave. During a duration of 10 years (2025 – 2035) a maximum bed level change of +1.0m is expected. As the height of the rock berms range between 1.13m (Cable-only) and 2.12 (CSS) silt accumulation on the ReefBooster tiles is unlikely at this location.

4.2 Recommendations

The following recommendations result from this desk study:

1. Validity of the HaSPro calculation methodology could be verified with a physical model test campaign estimating specific mobility limits for the ReefBooster tiles.
2. In addition to the hydraulic stability limits, the ecological stability limits could be defined such that the hydraulic stability can be linked to the ecological functionality of the tiles. It is likely that some movement or local displacement of the tiles does not compromise their ecological functionality.
3. Instead of a fixed size and density of the tiles, it could be considered to estimate the required size and density for a stable solution at a specific location.
4. It could be considered to define different extreme design events to estimate the stability of the tiles, considering the trade-off between the temporary nature of the ReefBooster tiles and the sensitivity of working with living oysters.

5 References

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A Cumulative distribution functions Monte Carlo simulation

Nearshore location

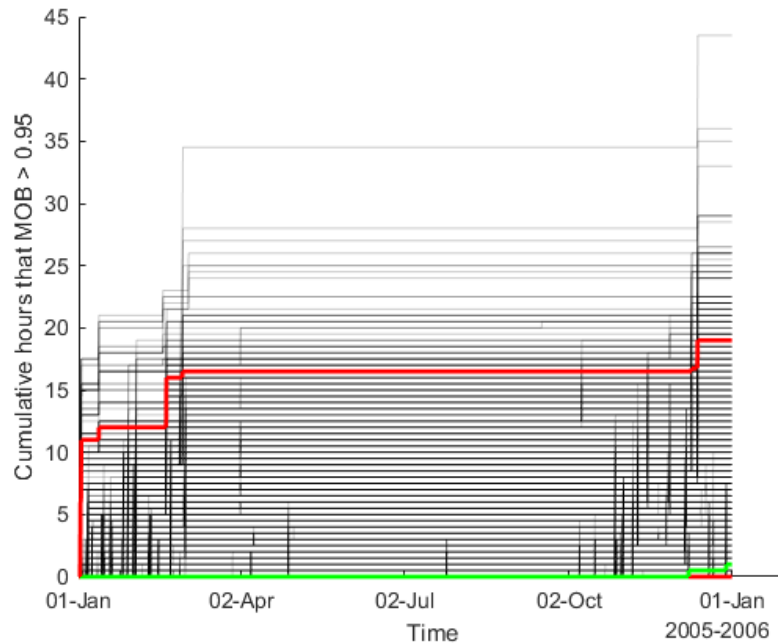


Figure A.1 Density 2000 kg/m³, seabed level

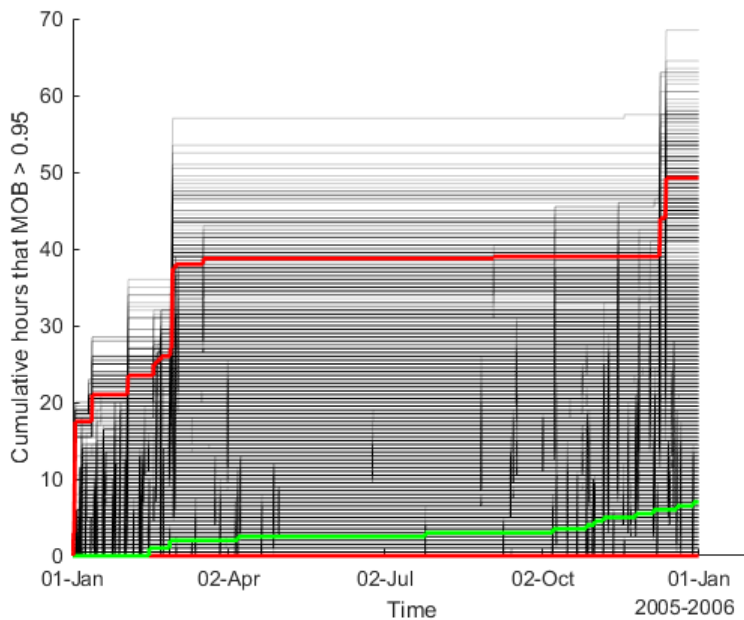


Figure A.2 Density 2000 kg/m³, cable-only rock berm

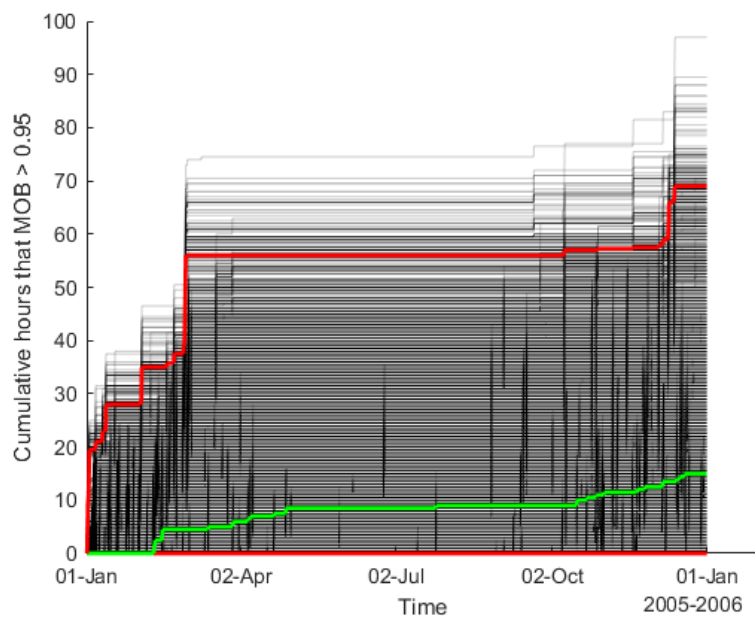


Figure A.3 Density 2000 kg/m³, CSS rock berm level

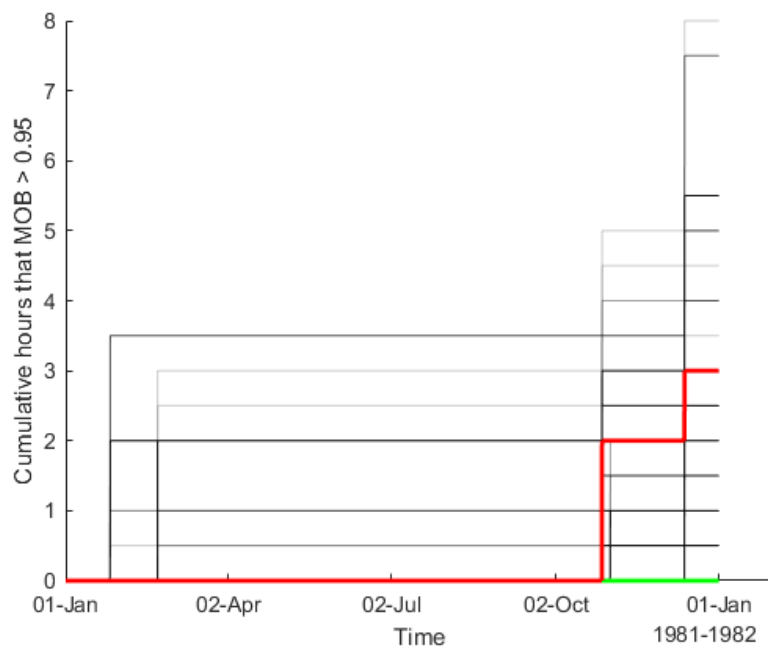


Figure A.4 Density 2330 kg/m³, seabed level

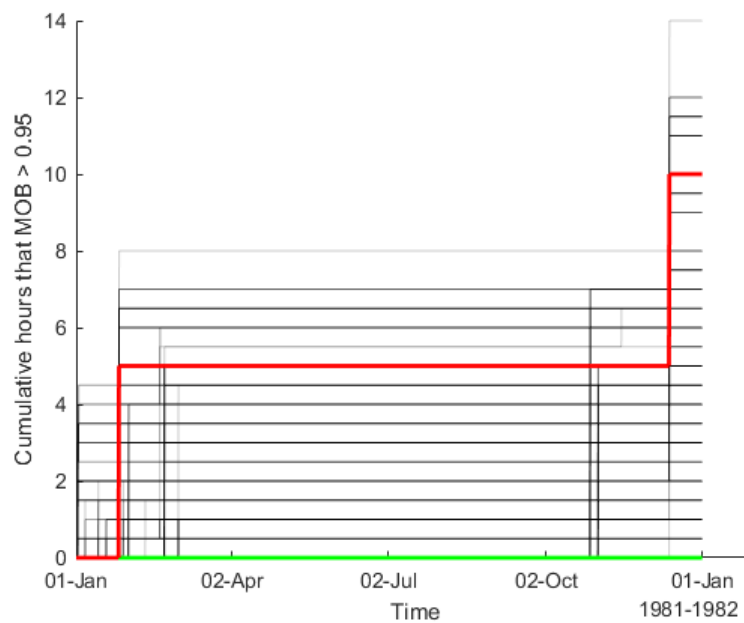


Figure A.5 Density 2330 kg/m³, cable-only rock berm

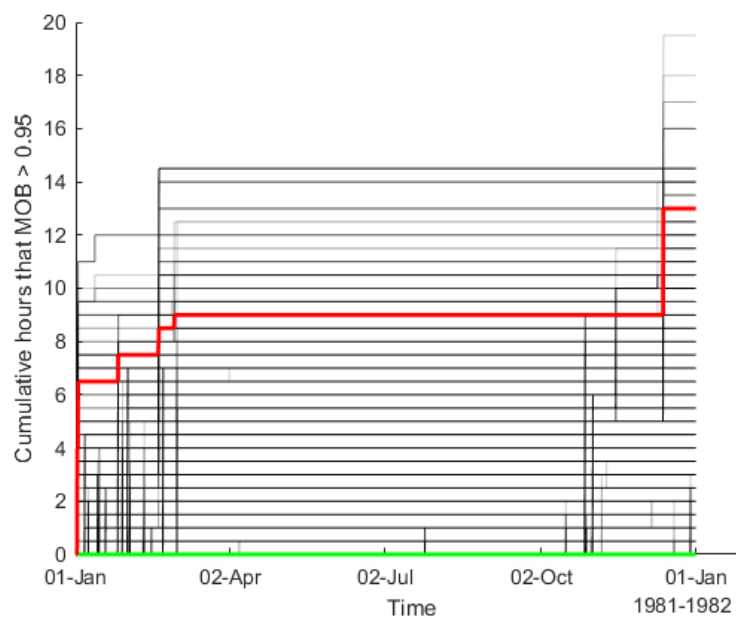


Figure A.6 Density 2330 kg/m³, CSS rock berm level

Offshore location

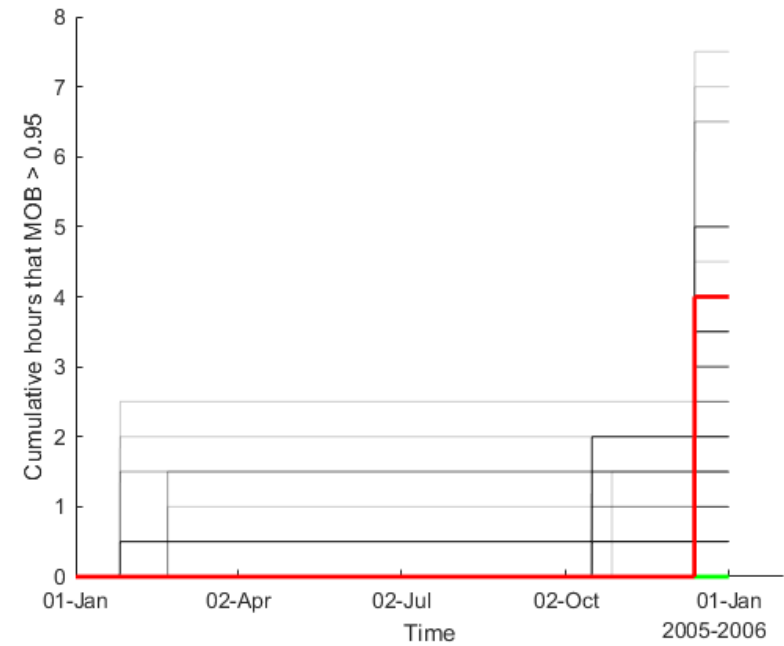


Figure A.7 Density 2000 kg/m³, seabed level

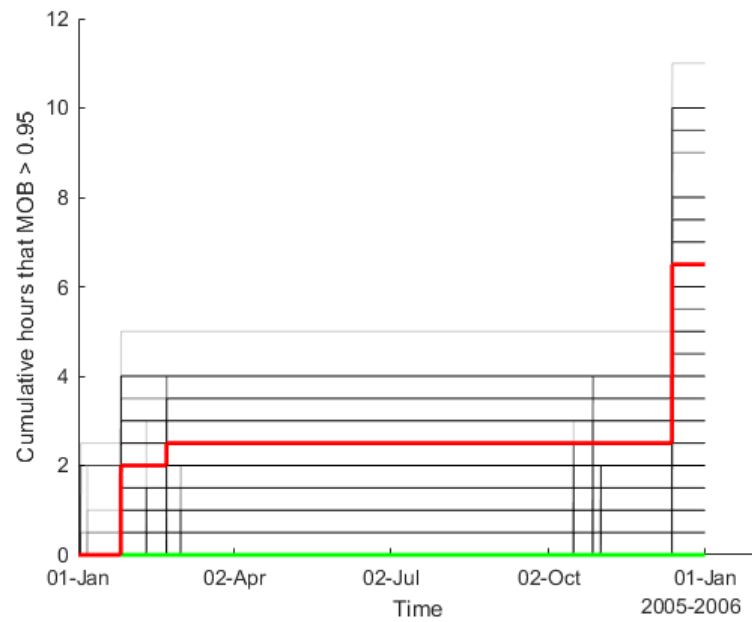


Figure A.8 Density 2000 kg/m³, cable-only rock berm

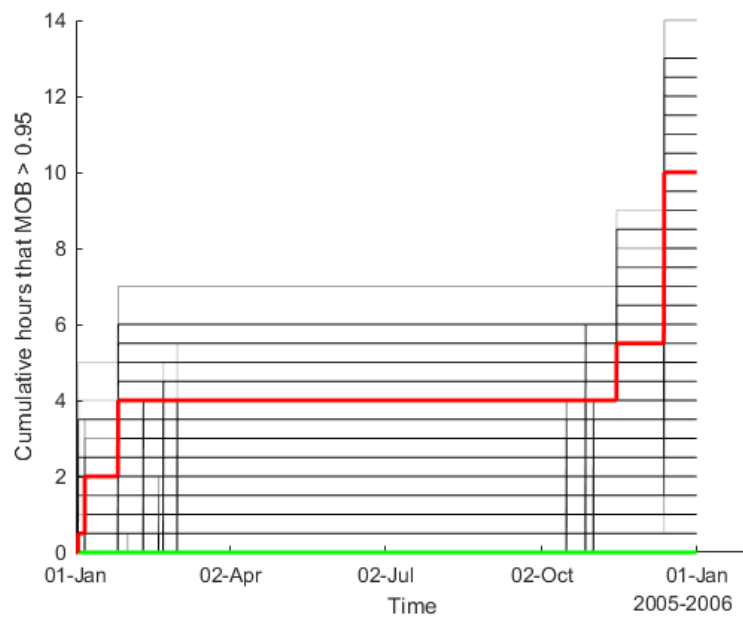


Figure A.9 Density 2000 kg/m³, CSS rock berm level

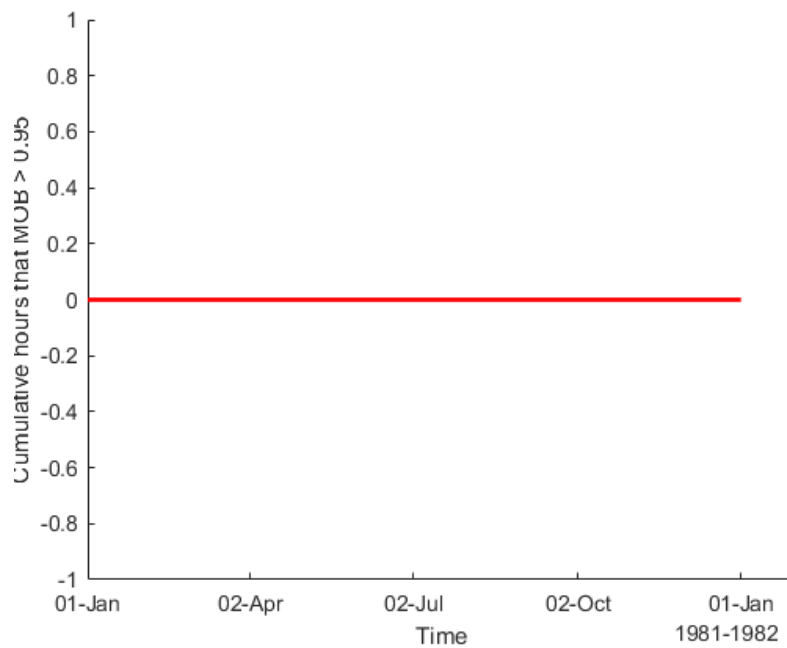


Figure A.10 Density 2330 kg/m³, seabed level

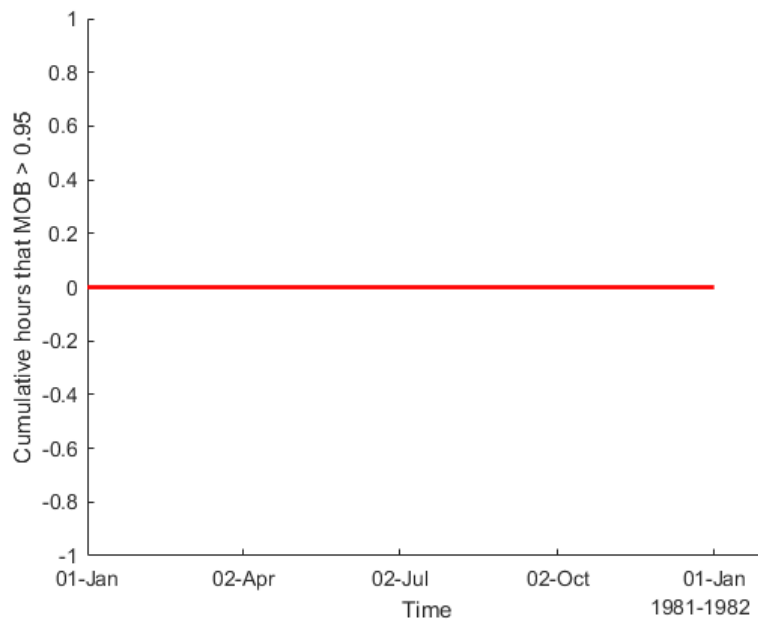


Figure A.11 Density 2330 kg/m³, cable-only rock berm

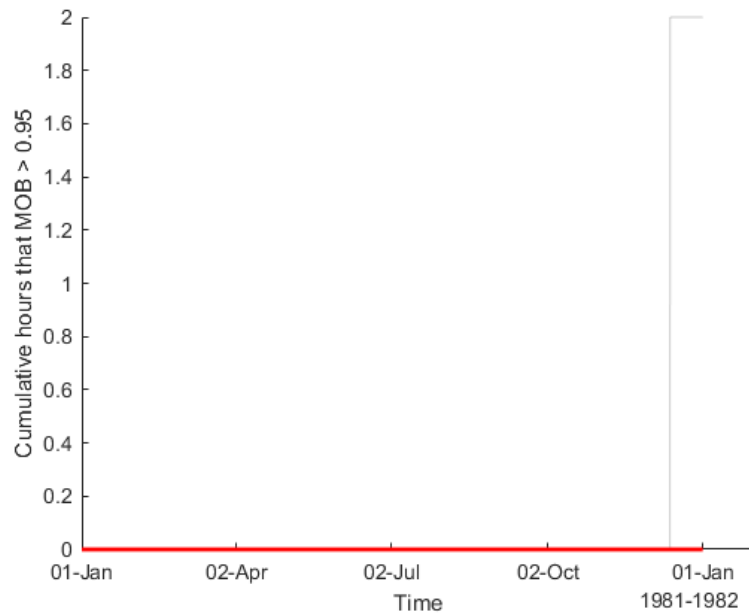


Figure A.12 Density 2330 kg/m³, CSS rock berm level

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